Prefabs

GameSession
Transform

EnemyFactory
Transform

GooberBoss(Enemy) Tag: Enemy
Transform
Animator
Sprite Renderer
Capsule Collider 2D
Rigidbody 2D
Box Collider 2D
Network Object
Network Transform
Client Network Animator

Goober(Enemy) Tag: Enemy
Transform
Animator
Sprite Renderer
Capsule Collider 2D
Rigidbody 2D
Box Collider 2D
Network Object
Network Transform

Ginger(Player) Tag: Player
Transform
Animator
Sprite Renderer
Capsule Collider 2D
Rigidbody 2D
Player Input
Box Collider 2D
Network Object
Client Network Transform
Network Rigidbody 2D
Client Network Animator
AttackPoint (Child)

FallDetector Tag: FallDetector
Transform
Box Collider 2D

Checkpoint Tag: Checkpoint
Transform
Sprite Renderer
Box Collider 2D

Manager
DataPersistenceManager(Child)

Akai

Controller

UIController

GameSession: MonoBehaviour, IDataPersistence
- playerMaxP: int
- score: int
- healthPointsText: TextMeshProUGU
- scoreText: TextMeshProUGUI

- Awake(): void
+ LoadData(GameData): void
+ SaveData(ref GameData): void
- Start(): void
+ AddScore(int): void
- UpdateHP(int): void
- ResetHP(): void
- SetHP(Transform): void

MapUIController: MonoBehaviour
- myCanvas: Canvas

- Start(): void
- Update(): void
- ShowCanvas(): void
- HideCanvas(): void

NetworkUIController: MonoBehaviour
- debugText: dF
- startServerButton: Button
- startHostButton: Button
- startClientButton: Button
- ipAddressText: TextMeshProUGUI
- ipInput: TMP_InputField
- transport: UnityTransport
- currentID: ulong

- Start(): void
+ StartHost(): void
+ StartServer(): void
+ StartClient(): void
- StartClientCoroutine():IEnumerator
+ GetLocalIPAddress(): String
+ SetIPAddress(string):void
- HideConnectionUI():void
- FindPlayerInstanceByID(ulong): GameObject
- waiter(): IEnumerator

CameraController: MonoBehaviour
- cinemaCamera: CinemachineVirtualCamera

- Start(): void
- FollowWithCamera(Transform): void

Logger

CoinController: MonoBehaviour
- pointsForCoinPickup: int

- OnTriggerEnter2D(Collider2D): void

EventHolder: MonoBehaviour
+ instance: static EventHolder

- Awake(): void
+ OnClientSpawnedEvent: Action<Transform>
+ OnPlayerDamageEvent: Action<int>
+ OnPlayerDamage(int): void
+ OnPlayerDiedEvent: Action
+ OnPlayerDied(): void

CharacterController

ICharacterController: Interface
- Die(): void
- TakeDamage(int): void

Enemy

EnemyController: NetworkBehaviour, ICharacterController
enemyData: EnemyData
myRigidbody: Rigidbody2D
animator: Animator

- Start(): void
- Update(): void
+ TakeDamage(int): void
+ Die(): void
- OnTriggerExit2D(Collision2D): void
FlipEnemyFacing(): virtual void

BigEnemyController: EnemyController, ICharacterController
- bigEnemyData: BigEnemyData
- enemyFactory: EnemyFactory
- smallEnemyCounter: int

- Start():void
+ TakeDamage(int): new void
- SpawnSmallEnemy(): IEnumerator

SmallEnemy: EnemyController

EnemyFactory: MonoBehaviour
+ instance: static EnemyFactory
- mobPrefab: GameObject
- bossPrefab: GameObject

- Start(): void
- Awake(): void
+ CreateEnemy(EnemyType, float, float): void
- GetPrefabType(EnemyType): GameObject
- GetRandomSpawnPosition(float, float): Vector2
- IsSpawnPositionValid(Vector2): bool
+ EnemyType: enum

Player

PlayerController: NetworkBehaviour
- playerData: PlayerData
- playerSettings: PlayerSettings
- attackPoint: Transform
- enemyLayers: LayerMask
- myRigidbody2D: Rigidbody2D
- myAnimator: Animator
- myFeetCollider: BoxCollider2D
- jumpCount: int
- canTakeDamage: bool
- isAlive: bool
- isTouchingCheckpoint: bool
- normalGravity: float
- isStomping: bool
+ GetMaxHP(): int

- Start(): void
- Update(): void
- OnStomp(InputValue): void
- OnCollisionEnter2D(Collision2D): voi
- Stomp(): IEnumerator
- CheckStomping(): bool
- StompEnemy(Collision2D): void
- BouncePlayer(Vector2): void
- Attack(): void
- OnDrawGizmosSelected(): void
- Jump(): void
- Dash(): void
- DoDoubleJump(): bool
- Run(float): void
- DoDash(): IEnumerator
- FlipSprite() void
- Die(): void
- Respawn(): void
- CheckToTakeDamage(): void
- TakeDamage(): void
- WaitToTakeDamage(): void
- OnTriggerEnter2D(Collider2D): void
- OnTriggerExit2D(Collider2D): void

Persistence

SaveSystem

DataPersistenceManager: MonoBehaviour
- fileName: String
- gameData: GameData
- dataPersistenceObjects: list<IDataPersistence>
- dataHandler: FileDataHandler
+ instance: static DataPersistenceManager

- Awake(): void
- Start(): void
+ NewGame(): void
+ LoadGame(): void
+ SaveGame(): void
- OnApplicationQuit(): void
- FindAllDataPersistenceObjects(): list dataPersistenceObject

FileDataHandler
- dataDirPath: String
- dataFileName: String

+ FileDataHandler(String, String)
+ Load(): GameData
+ Save(GameData): void

IDataPersistence: Interface
- LoadData(GameData): void
- SaveData(ref GameData): void

Data

Player

PlayerData
- canDoubleJump: bool
- canDash: bool
- canStomp: bool
- maxHealthPoints: int
- currentHealthPoints: int
- respawnPoint: Vector3

+ CanDoubleJump(get, set): bool
+ CanDash(get, set): bool
+ CanStomp(get, set): bool
+ MaxHealthPoints(get, set): int
+ CurrentHealthPoints(get, set): int
+ RespawnPoint(get, set): Vector3

PlayerSettings
- runSpeed: float
- jumpSpeed: float
- dashSpeed: float
- dashJumpSpeed: float
- dashDuration: float
- timeBetweenDash: float
- attackRange: float
- attackDamage: int
- stompGravity: float
- stompPenalty: float
- stompDamage: int
- bounceForce: float

+ RunSpeed(get, set): float
+ JumpSpeed(get, set): float
+ DashSpeed(get, set): float
+ DashJumpSpeed(get, set): float
+ DashDuration(get, set): float
+ TimeBetweenDash(get, set): float
+ AttackRange(get, set): float
+ AttackDamage(get, set): int
+ StompGravity(get, set): float
+ StompPenalty(get, set): float
+ StompDamage(get, set): int
+ BounceForce(get, set): float

Enemy

BigEnemyData
- smallEnemySpawnRate: float
- smallEnemySpawnCount: int

+ BigEnemyData(float, int)
+ SmallEnemySpawnRate(get, set): floa
+ SmallEnemySpawnCount(get, set): int

EnemyData
- moveSpeed: float
- maxHealth: int
- currentHealth: int

+ EnemyData(float, int)
+ CurrentHealth(get, set): int
+ MoveSpeed(get, set): float
+ MaxHealth(get, set): int

GameData
+ score: int

+ GameData()

Presentation

Prefabs

CanvasMap
Rect Transform
Canvas
Canvas Scaler
Graphic Raycaster

NetworkUI
Rect Transform
Canvas
Canvas Scaler
Graphic Raycaster

Main Camera
Transform
Camera
Audio Listener
CinemachineBrain

Follow Camera
Transform
CinemachineVirtualCamera
Cinemachine Confiner

Map Camera
Transform
Camera

Coin
Transform
Animator
Sprite Renderer
Circle Collider 2D

EventHolder
Transform